

THE HOPECORE PROJECT

Game Development Software Guide

Engines · Hosting · Audio · Art · Level Design · VCS · Distribution · Analytics — 2026

■ GAME ENGINES — Fully Free / Open Source

Godot Engine

Fully open-source 2D/3D engine. GDScript + C#. No royalties ever. Rapidly growing indie favourite.

100% Free

Win

Mac

Linux

GDevelop

No-code 2D/3D engine with event-based logic. Free forever plan. Exports to Web, Android, iOS, PC.

Free (core)

Win

Mac

Browser

Cocos Creator

Open-source 2D-first engine for mobile. Cross-platform to Web, Android, iOS. Supports Lua and JS.

Open Source

Win

Mac

Linux

Phaser (HTML5)

Best open-source HTML5 game framework. Canvas + WebGL. Ideal for browser games and game jams.

Open Source

Browser

Node.js

LibGDX

Java-based cross-platform framework. Desktop, Android, iOS, Web. Mature, performant, beloved by Java devs.

Open Source

Win

Mac

Linux

Defold

Free, source-available 2D engine by King. Lua scripting, tiny footprint. Built for mobile and web games.

Free

Win

Mac

Linux

■ GAME ENGINES — Free with Limits / Freemium

Unity

Personal tier free up to \$200K revenue. Powers ~48% of mobile games. Runtime fee cancelled Sept 2024. Pro required above \$200K.

Free
(Personal)

Win

Mac

Linux

Pro: ~\$2,040/yr per seat

Unreal Engine 5

Free to use; 5% royalty on revenue over \$1M per title. Photorealistic Nanite/Lumen, Blueprint visual scripting. AAA standard.

Free (5%
royalty)

Win

Mac

Linux

Free until \$1M revenue/title

GameMaker

Free for non-commercial use. Optimised for 2D. Lightweight, beginner-friendly. Console export requires paid subscription.

Free
(Non-commercial)

Win

Mac

Indie: \$99/yr | Enterprise: \$799/yr

Construct 3

Event-driven 2D/2.5D engine with new 3D support (2026). Free trial limited by events and exports. No coding required.

Free (Limited trial)

Browser

Win

Personal: \$97.69/yr | Business: \$358/yr

O3DE (Open 3D Engine)

Open-source successor to Amazon Lumberyard. AAA-grade, community-driven, with native AWS integration.

Free / Open Source

Win

Linux

RPG Maker MZ

Purpose-built for 2D JRPGs. Free trial available. Huge asset marketplace. Best for solo devs making story games.

Free Trial

Win

Mac

~\$79.99 one-time

■■ GAME HOSTING & BACKEND SERVICES

Free / Open-Source / Self-Hostable

Nakama (Heroic Labs)

Open-source game server backend. Auth, real-time multiplayer, leaderboards, social, cloud saves. Self-hostable for free.

Open Source

Self-hosted

Cloud

Managed: paid by cloud tier

Firebase (Google)

Realtime database, auth, cloud functions, crash reporting, analytics. Generous free Spark plan. Works with any engine.

Free (Spark plan)

Cloud

Blaze: pay-as-you-go

Supabase

Open-source Firebase alternative. Postgres DB, auth, storage, edge functions. Free tier for small projects. Self-hostable.

Free tier

Cloud

Self-hosted

Pro: \$25/mo

Freemium Backends

Microsoft PlayFab

Azure-backed backend. Free in dev up to 10K players. Multiplayer servers, LiveOps, analytics, matchmaking. Powers Halo & Sea of Thieves.

Free (Dev/10K players)

Cloud

Standard: pay-as-you-go | Enterprise: \$10K+/mo

LootLocker

Game backend with player management, leaderboards, file storage, asset delivery. Free indie tier. SDKs for Unity, Unreal, Godot.

Free tier

Cloud

Paid tiers scale with MAU

Photon Engine (PUN/Fusion)

Industry-standard real-time multiplayer networking. 20 CCU free forever. PUN for Unity, Fusion for competitive titles.

Free (20 CCU)

Cloud

From \$95/mo (100 CCU)

GameSparks (AWS)

AWS-owned backend. Free during dev up to 100 CCU. Leaderboards, matchmaking, economies, analytics.

Free (100 CCU dev)

Cloud

Launch: \$299/mo

Dedicated Game Server Hosting

Amazon GameLift

AWS-managed dedicated game server hosting. Auto-scaling, global regions, matchmaking. Pay-per-use.

Paid

Cloud / AWS

Pay-as-you-go (VM hours + egress)

Edgegap

Latency-aware game server orchestration across 600+ PoPs. Auto-scales globally. Free tier available. Unity Netcode integration.

Free tier

Cloud

Pay per compute used

Hathora

Cloud-native game server platform. Auto-scales to zero when idle. Pay only when players are connected.

Free tier

Cloud

Pay per active server hour

■ GAME AUDIO TOOLS

FMOD Studio

Industry-standard adaptive audio middleware. Event-based sound design, real-time mixing, Unity/Unreal integration.

Free
(Revenue
limit)

Win

Mac

Linux

Paid: \$2,000/title above threshold

Wwise (Audiokinetic)

Gold standard — 70% of AAA titles. Free up to 200 sound files / non-commercial. Scales to enterprise licensing.

Free (200
sounds)

Win

Mac

Linux

Commercial: revenue-based pricing

Audacity

Free open-source audio editor. Recording, trimming, noise removal, effects. Perfect for SFX and voice overs.

Open Source

Win

Mac

Linux

BFXR / SFXR

Free 8-bit retro sound effect generator. Procedurally generates SFX. Browser-based, zero setup. Beloved for game jams.

Free

Browser

Win

Reaper

Full-featured DAW used in game audio production. Affordable indie license. Highly scriptable, low CPU footprint.

60-day trial

Win

Mac

Linux

Indie license: \$60 one-time

FL Studio

Popular DAW for composing game music. One-time purchase with free lifetime updates. Great for electronic game soundtracks.

Demo (no save)

Win

Mac

Producer: \$199 one-time

■ 2D ART, SPRITES & ANIMATION

Aseprite

Go-to pixel art and animation tool. Frame-by-frame, onion skinning, tilemap support. Affordable one-time purchase.

Paid

Win

Mac

Linux

\$19.99 one-time

Libresprite

Free open-source fork of old Aseprite. Full pixel art editor with animation timeline. Best free Aseprite alternative.

Open Source

Win

Mac

Linux

Spine 2D

Industry standard for 2D skeletal animation in games. Runtime support for Unity/Unreal/Godot. Used in major indie titles.

Paid

Win

Mac

Linux

Essential: \$69 | Pro: \$299

DragonBones Pro

Free alternative to Spine. Skeletal 2D animation, export to Unity, Egret, Cocos. Good for indie budget constraints.

Free

Win

Mac

Toon Boom Harmony

Professional frame-by-frame and cut-out animation. Behind Rick and Morty and countless game cutscenes.

Paid

Win

Mac

Essentials: \$25/mo | Premium: \$100/mo

Piskel

Free browser-based pixel art and animation editor. Simple, fast, no install. Good for quick sprites and jam assets.

Free

Browser

Win

■■ LEVEL DESIGN & MAP TOOLS

Tiled Map Editor

Free open-source 2D tilemap editor. Export to JSON, XML, TMX. Supported by nearly every 2D engine. The standard for 2D level design.

Open Source

Win

Mac

Linux

LDtk (Level Designer Toolkit)

Modern level editor with smart auto-layering and world maps. Free and open-source. Created by the Celeste devs.

Open Source

Win

Mac

Linux

ProBuilder (Unity)

In-editor 3D level prototyping built into Unity. Grey-box environments, UV unwrapping, quick level sketches.

Free (Unity Package)

Win

Mac

World Machine

Node-based procedural terrain generator. Exports heightmaps and textures for any engine. Used for large open worlds.

Free (Limited res)

Win

Pro: \$149 one-time

VERSION CONTROL & ASSET MGMT

Git + GitHub / GitLab

Standard source control for game code. Free for public and private repos. Git LFS handles large art assets.

Free

Win

Mac

Linux

Perforce Helix Core

Industry VCS used by EA, Rockstar. Free up to 5 users / 20 workspaces. Excellent for large binary game assets.

Free (5 users)

Win

Mac

Linux

Paid: volume-based

Unity Version Control (Plastic SCM)

Git-like VCS built for game art pipelines. Branching, locking, visual diff of scenes. Integrated with Unity.

Free (3 users)

Win

Mac

Paid tiers available

Anchorpoint

Git-based asset management for game art teams. File locking, previews, version history, review tools for Unity/Unreal pipelines.

Free (small teams)

Win

Mac

Team plans available

GAME DISTRIBUTION & PUBLISHING

Steam (Valve)

Largest PC game store. \$100 one-time app fee. 70/30 revenue split (88% after \$10M). Huge install base, workshop, achievements.

\$100 app fee

PC

Mac

Linux

30% revenue cut

itch.io

Indie-friendly open marketplace. Free to publish. Developer sets their own revenue split (can be 100%). Best for indie/jam games.

Free to publish

Web

Desktop

You choose the cut

Epic Games Store

88/12 revenue split (dev keeps 88%). Growing PC store. Self-publishing available. Free games program grows user base.

Free to submit

PC

12% revenue cut

Google Play / Apple App Store

Primary mobile distribution. Google Play: \$25 one-time. App Store: \$99/yr. Both take 15-30% cut.

Paid registration

Android

iOS

GP: \$25 once | AS: \$99/yr

GameJolt

Free game hosting and community for indie games. No publishing fees. Built-in jam support, achievements, follower system.

Free

Web

Newgrounds

Veteran web platform for HTML5 games. Free to publish. Community ratings and revenue sharing for popular titles.

Free

Browser

■ GAME ANALYTICS

Unity Analytics

Built into Unity. Funnel analysis, event tracking, A/B testing, player segmentation. Free with Unity Personal.

Free (with Unity)

Cloud

GameAnalytics

Free analytics platform. Session tracking, retention, IAP funnels, error tracking. SDKs for Unity, Unreal, Godot, Cocos.

Free

Cloud

Amplitude

Product analytics with powerful segmentation. Widely used by studios for LiveOps and engagement metrics.

Free (10M events/mo)

Cloud

Growth: from \$49/mo

DeltaDNA (Unity)

Game-specific data platform (now part of Unity). Personalisation engine, smart campaigns, player behaviour analysis.

Free tier

Cloud

Green = fully free/open source • Orange = free tier with restrictions • Pink = paid/commercial • Prices approximate and subject to change.